Do Machines Listen Like Humans?
A Temporal Benchmark for Phonological Competition in End-to-End ASR

Anonymous submission to Interspeech 2025

Abstract

Human speech perception is incremental, with listeners activating and suppressing potential words in real time as they hear speech. Some researchers have explored using modern Automatic Speech Recognition (ASR) models as potential models of human speech recognition (HSR). Here we ask whether ASR models process speech with human-like incremental dynamics, focusing on the time course of lexical activation and phonological competition. We propose a benchmark to quantitatively compare the time course of HSR and ASR, using a point-wise comparison of internal model activation trajectories against human eyetracking data. We define metrics to trace word activation profiles across various architectures. Our results reveal a clear dissociation: Causal models successfully replicate the hallmark human pattern of early competition between onset competitors, and later, weaker competition between words that rhyme, whereas non-causal models with 'look-ahead' mechanisms fail to capture these temporal dynamics. These findings demonstrate that architectural constraints are critical for developing speech systems that process language in a human-like manner, and suggests caution is warranted in considering ASR or other large language models as models of human language processing[1].

Index Terms: speech recognition, human-computer interaction, computational paralinguistics

1. Introduction

Human speech perception is inherently incremental; listeners do not wait for the offset of a word to begin processing its meaning. Instead, they continuously accumulate evidence in support of or against lexical candidates in real time as the acoustic signal unfolds. A foundational method for estimating these internal dynamics in HSR is the Visual-World Paradigm (VWP), which tracks eye movements toward objects in a display to reveal how lexical activation and competition fluctuate over time [1]. A typical VWP display and fixation patterns (averaged over many items and subjects) from a seminal study are illustrated in Figure 1.

Historically, this incremental process has been explained through several influential psycholinguistic models. The Cohort model [2] posits that word recognition begins with a "cohort" of all words matching the initial acoustic input (the onset), which is then winnowed down as more information arrives. In contrast, the TRACE model [3] suggests a connectionist framework where competition is more fluid and interactive, allowing for the activation of "rhymes" (words that share endings but differ

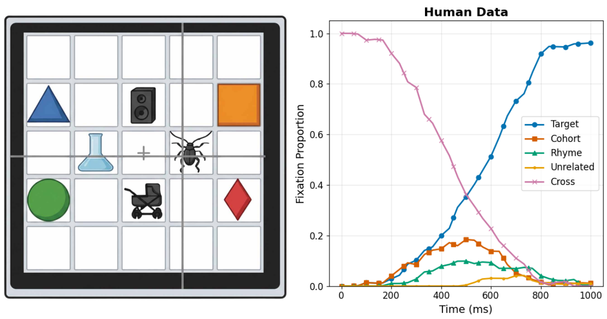

Figure 1: *Left. Dynamics of phonological competition in humans and models. Listeners followed spoken instructions (e.g., 'click on the beaker') to interact with simple displays (left), and tracked their eye movements as they did so (right). Displays included targets (e.g., beaker), cohorts (onset competitors, e.g., beetle), rhymes (e.g., speaker), and unrelated baseline items (e.g., carriage). Cross is defined as the complement of the total activation of all other objects. Fixation proportions over time are hypothesized to relate to internal lexical activation and competition.*

at onset) even without an initial onset match. More recently, the EARSHOT model [4] has bridged these theories with modern computational approaches, using end-to-end neural networks to simulate the complexities of spoken word recognition.

These theories predict a specific time course of lexical activation, which has been confirmed by human behavioral data [1]. In the VWP, cohorts that share the target's onset (e.g., beetle, when the target is beaker) compete for activation early in the signal, while rhymes that mismatch at onset but share subsequent phonemes (e.g., speaker) typically peak in activation later. Unrelated distractors with no significant phonological or semantic similarity to the target provide a baseline for estimating competition [1]. These well-documented dynamics offer a precise quantitative benchmark of incremental human word recognition.

Neural network models serve as the foundation of modern ASR and have significantly advanced speech-to-text performance. Several research teams have explored using various ASR/large language models as models of the neurobiology of HSR [5, 6, 7, 8, 9, 10, 11]. While significant correlations between internal model activity and measures of neural activity in humans listening to the same speech inputs have been reported, it remains largely unexplored whether these end-to-end models process speech with human-like incremental dynamics. If they are unable to emulate the time course of HSR, this would challenge their validity as neurobiological models.

Notably, many high-performing ASR systems rely on *non-causal architectures* with "look-ahead" mechanisms that allow

[1]Code is available in `https://anonymous.4open.science/r/HTP-benchmark-15D4/README`

the model to utilize future context. We hypothesize that this architectural choice may lead to lexical activations that fundamentally deviate from the causal, past-to-present nature of human hearing. This distinction is important because, while such systems may be well optimized for the practical goal of recognizing speech, they are not necessarily well suited as psychological models of HSR—and we view the latter as an important goal in its own right. Therefore, we propose a novel benchmark to evaluate whether a variety of end-to-end ASR systems exhibit human-like patterns of phonological competition. We use HSR timecourse estimates from the VWP as a baseline to probe model states over time and trace word activation profiles across diverse architectures, including LSTMs, CNNs, and Transformers [12, 13, 14]. Our objective is to determine if architectural constraints—specifically temporal causality—are necessary for machines to "listen" and resolve lexical ambiguity in a manner compatible with human cognitive processes. We further examine how this constraint manifests in two prominent classese of modern ASR models[2]: connectionist temporal classification [18] (CTC)-based models and encoder-decoder models with attention. By incorporating models that vary in scale (e.g., the large-scale wav2vec2.0 [19]), loss criterion, and architectural details, we are able to test our hypothesis across a wider and more ecologically valid range of current approaches.

2. Method

We follow the word-level training approach from EARSHOT [4] but use centered 300-dimensional word2vec embeddings[3] as target representations instead of binary targets. The model is trained with Mean Squared Error (MSE) loss, which encourages the output to match the target embedding at each time frame. Rather than replicating standard ASR training with CTC or massive datasets, this setup serves as a controlled experiment with a biologically plausible learning object: The model learns to activate a distributed pattern that resembles neural representations of words.

To systematically evaluate whether end-to-end ASR models exhibit human-like incremental word competition, we selected a range of neural network architectures that vary in their temporal processing mechanisms. These models were chosen to reflect both foundational designs commonly used in speech recognition and more recent advances in sequence modeling. During evaluation, we measure the cosine similarity between the model's output, the target's word2vec embedding, and the embeddings of potential phonological competitors at each time step. This gives us a time course of word activation for each model, enabling direct comparison with human behavioral trajectories.

In term of foundation models, we evaluate—wav2vec 2.0[4] , HuBERT[5] [20], and Whisper[6]—using their pre-trained versions without fine-tuning. To derive word activation estimates, we compute word-level probabilities from CTC alignment paths (for wav2vec 2.0 and HuBERT) and from attention-weighted token probabilities (for Whisper). This enables comparison of activation dynamics across diverse training paradigms within a unified framework grounded in lexical competition.

2.1. Dataset

A total of 1,533 uninflected English words were randomly selected from a lexicon, with lengths ranging from 1 to 16 phonemes (mean = 6.09). Pronunciations of all 1,533 words were generated by six talkers from the Apple application "Say" and one human speaker, resulting in $7 \times 1,533$ audio files in total. Prior to comparison with human fixation proportions, we inserted 200 ms silence at word onset to approximate the lag between signal information and changes in human fixation proportions [1]. Each audio file was converted into a spectrogram $\mathbf{x}_i \in X$, where X denotes the set of all spectrograms. Each spectrogram $\mathbf{x}_i$ is represented as a matrix with 256 frequency channels and a 10 ms time step resolution, derived from an audio signal sampled at 16k Hz. For each word, a word2vec embedding served as the semantic target. All target vectors $\mathbf{s}_i \in \mathcal{S}$ were centered (i.e., the mean vector of the set was subtracted). For each of the 7 talkers, approximately $1/7$ of the words were randomly selected and reserved as test items. The remaining 1,314 words × 7 talkers (9,198 items) formed the training set.

2.2. Network architecture

The baseline model comprises a single LSTM layer followed by a feed-forward layer [4]. For a given word i, the encoder $r : X \rightarrow \mathcal{H}$ processes the spectrogram $\mathbf{x}^i \in X$ through the LSTM layer, producing a sequence of latent speech representations $\mathbf{h}_{t=1}^{T}$. These hidden states are then passed through the feed-forward layer $f : \mathcal{H} \rightarrow \mathcal{W}$ to obtain a word-level representation. Specifically, we take the final hidden state $\mathbf{h}_T$ and compute:

$$\{\mathbf{w}_t\}_{t=1}^{T} = \tanh\big(f(h(\mathbf{x}_1, \ldots, \mathbf{x}_T))\big),$$

where $\tanh$ denotes the hyperbolic tangent activation function[7]. The model is trained with MSE between the predicted vector $\mathbf{w}_t$ and the target semantic vector $\mathbf{s}_i$.

Table 1: *Parameter counts, training and test accuracy.*

Model	Parameters	Train Acc	Test Acc
Baseline	1,730,860	0.95	0.41
Causal-2L-LSTM	1,657,900	0.95	0.65
Causal-RCNN	1,659,692	0.98	0.72
Causal-CNN	1,869,868	0.81	0.52
Causal-Transformer	1,660,204	0.67	0.38
Causal-ConvTransformer	1,857,580	0.99	0.15
2L-BiLSTM	1,601,836	0.77	0.66
RCNN	1,659,692	0.99	0.84
CNN	1,869,868	0.93	0.81
Transformer	1,660,204	0.89	0.62
ConvTransformer	1,857,580	0.99	0.84

2.2.1. *Models*

To assess the impact of temporal architecture, we compared causal and non-causal variants of several networks, all with roughly comparable parameter counts (see Table 1).

The 5 causal models, constrained to process speech incrementally, were a Baseline single-layer LSTM, a deeper 2L-LSTM, a CausalCNN using only 1D convolutions, a Causal-RCNN combining a 1D CNN with a unidirectional LSTM, and

[2]We exclude RNN-Transducers [15] from this analysis because their inherent alignment delay—the time lag between audio input and text output—is an architectural feature that precludes a valid comparison with HSR timecourse [16, 17].

[3]https://dl.fbaipublicfiles.com/fasttext/vectors-wiki/wiki.en.vec

[4]https://huggingface.co/facebook/wav2vec2-base-960h

[5]https://huggingface.co/facebook/hubert-large-ls960-ft

[6]https://github.com/openai/whisper

[7]99% of values of the centered word2vec embeddings range from -1 to 1, and are normally distributed. Values out of range are clipped.

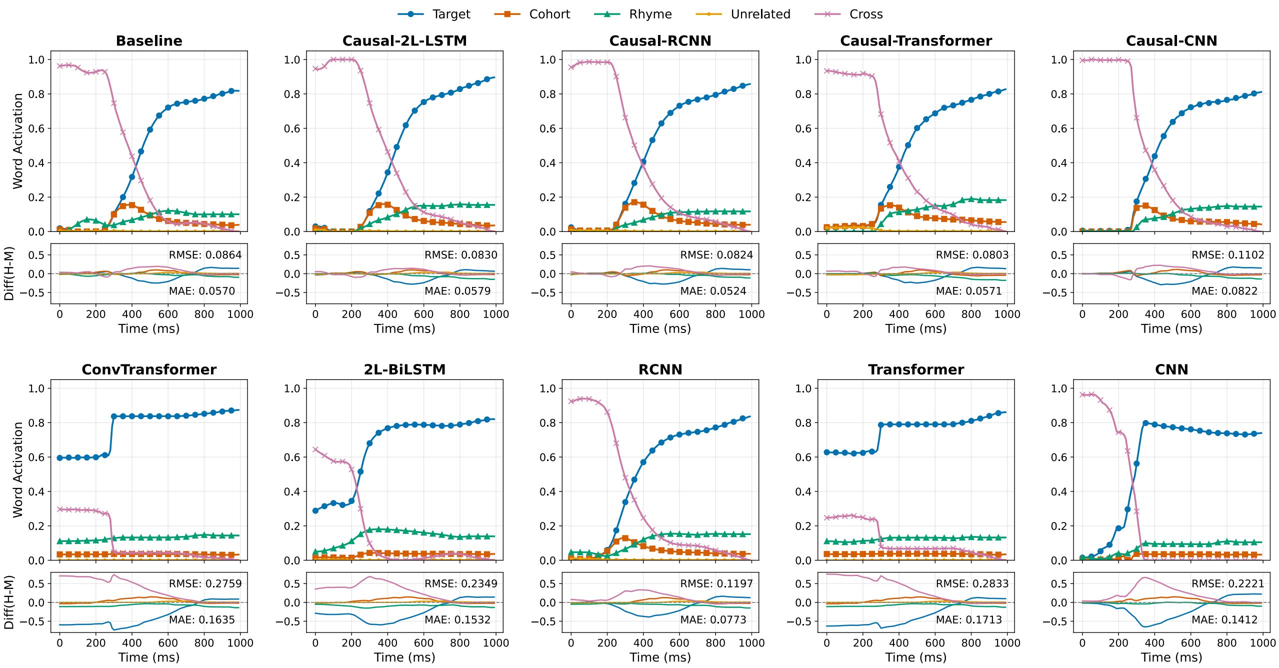

Figure 2: *Phonological competition time courses for each model (top row) and discrepancies between model simulations and human data (bottom row). For each model, the top panel shows the activation (cosine similarity) trajectories for target, cohort, rhyme, and unrelated words over time. The bottom panel displays the point-by-point deviation from human data, with RMSE and MAE summarized for each model.*

a CausalTransformer using causal self-attention. We also evaluated their non-causal counterparts (e.g., 2L-BiLSTM, RCNN, Transformer, ConvTransformer) to isolate the effect of future context on competition dynamics. Note that the RCNN model combines a non-causal CNN module (with a receptive field of 25 frames) with causal LSTM layers, allowing the model to look ahead 120 ms.

2.3. Metric

Word recognition accuracy was evaluated using the same highly conservative cosine-similarity threshold described in [4]. Let $cs_i(t)$ be the cosine similarity between the output vector $\mathbf{w}_t$ and the ith semantic vector $\mathbf{s}_i$ at time t, with i^* denoting the target index. Accuracy required that:

1. $cs_{i^*}(t) - \max_{i \neq i^*} cs_i(t) > 0.05$ for all t in an interval $\Delta t \geq 100$ ms (the target-output cosine exceeded all other word-output cosines by 0.05 for at least 100 ms).
2. $cs_{i^*}(t) > \max_{i \neq i^*} cs_i(t)$ for all t from the end of that interval until word offset (the target cosine remained the maximum cosine to word offset).

To compare the model's predictions against human data, we compute, for each target word, the mean cosine similarity over time for four competitor types: the target itself, cohorts (overlap in first two phonemes), rhymes (mismatch only in first phoneme), and unrelated words. Fixation cross probability is modeled as 1 minus the sum of these response probabilities at each time step.

These trajectories are then compared to human fixation proportions (Figure 1, right panel) using Root Mean Square Error (RMSE) and Mean Absolute Error (MAE) [21]. Metrics are computed point-wise for each competitor type and aggregated into global summary statistics.

3. Results and discussion

Table 1 shows word recognition accuracy. Non-causal models generally outperform their causal counterparts: RCNN and ConvTransformer reach 0.84 test accuracy, while the best-performing causal model reached 0.72 accuracy. Subsequent analyses focus on correctly recognized test words. Figure 2 displays activation time courses[8]. For each panel, the top graph shows activations (i.e. cosine similarity). The lower graph shows the raw difference between the upper graph and the human data replotted in the bottom right panel.

Causal models outperform non-causal ones, mirroring human incremental processing. Although non-causal models can achieve higher word recognition accuracy, they do not guarantee human-like behavior. Our results reveal a clear dissociation between causal and non-causal architectures in simulating human-like phonological competition (avg. RMSE/MAE: 0.07/0.05 vs. 0.22/0.14). This advantage aligns with the incremental nature of human speech processing: Causal models, constrained to access only past and present inputs, must resolve lexical competition in real time, like human listeners. They began activating words around 200 ms (that is, immediately after the 200 ms of silence inserted to reflect the 200-ms lag between phonetic details and fixation proportions in humans [1]) and produced cohort and rhyme trajectories that broadly matched expected patterns; cohort competitors (matching at word onset) showed early activation, while rhyme competitors (mismatching at onset but converging later) peaked later. While word2vec captures semantics, the fact that causal models replicate the temporal pattern suggests they are using phonology incrementally to constrain the semantic search.

Among causal architectures, baseline LSTM, 2L-LSTM,

[8]Due to space limitations and low accuracy, results for the Causal-ConvTransformer are not included.

and RCNN performed similarly, suggesting that recurrent connectivity is sufficient to approximate human-like dynamics under causal constraints. With the exception of the bidirectional RCNN, non-causal models showed (too) early activation of targets and rhymes, and much lower cohort activation than rhyme, indicating that access to future signal information alters the time course of lexical activation compared to HSR. Differences between the Transformer models and the bi2LSTM may indicate additional discrepancies introduced by self attention.

Note that the non-causal RCNN, with access to 120 ms of future context (unlike other non-causal models with global access), presenting an intriguing middle ground, approximated plausible competition patterns.

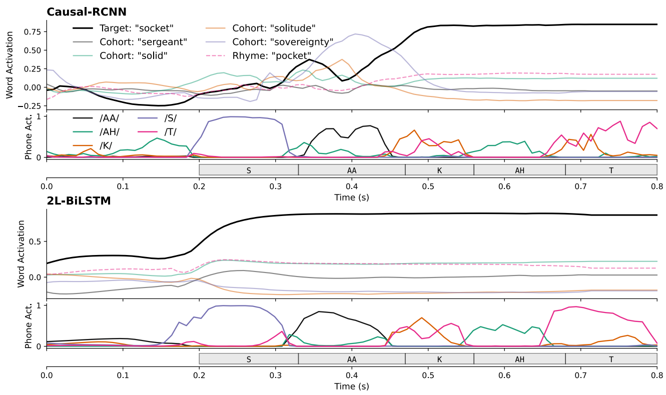

Figure 3: *Case study of target word "socket" comparing causal RCNN (top) and bidirectional 2L-LSTM (bottom). Upper panels show activation trajectories for target, cohort, and rhyme competitors over time. Lower panels show phoneme decoding from intermediate layers, with horizontal bars indicating phoneme boundaries from forced alignment.*

Bottom-up acoustic-phonetic input alone is insufficient. To examine whether internal representations track phonological competition over time, we decoded phoneme activations from intermediate representation of two models—applied to the first LSTM layer of the bidirectional 2L-LSTM and the CNN module of the causal RCNN. Both achieved high phoneme decoding accuracy ($> 60\%$), indicating that they encode fine-grained phonetic information. In the upper graphs in Figure 3, we show word activations and decoding results for the example stimulus, "socket." For the causal RCNN model, several cohort items were initially strongly activated, but the target word substantially suppressed them by the uniqueness point (phoneme /k/), while the rhyme competitor "pocket" rose above cohort activation thereafter, closely mirroring human visual-world dynamics. In contrast, the bidirectional 2L-LSTM model failed to reproduce this pattern, instead showing an immediate target advantage from stimulus onset (i.e., at the start of the leading silence), and then predicting similar cohort and rhyme activation. Although it also encoded phonetic information accurately, its access to future context allowed early "look-ahead," disrupting the natural time course of competition. Yet the time course of decoded internal phoneme representations (lower graphs) was quite similar, meaning an analysis tracking only phoneme decoding might overestimate the non-causal model's similarity to HSR.

Foundation ASR models diverge substantially from human-like incremental word recognition. Figure 4 shows the activation time courses of three foundational models. Among these, wav2vec 2.0 and HuBERT exhibited notably late acti-

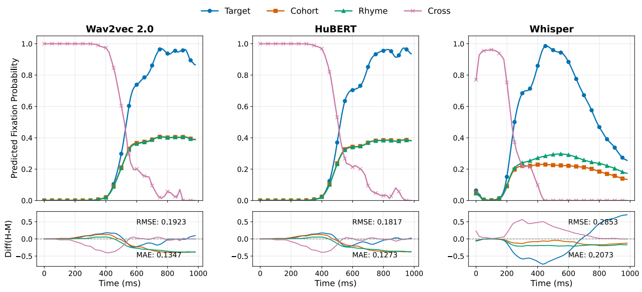

Figure 4: *Phonological competition time courses for foundation ASR models. Transformed using the Luce Choice Rule with $k = 6$ [1, 21].*

vation ($\sim 400ms$). This delay stems from CTC-based training, which aligns probability mass narrowly at specific frames rather than distributing it across the full duration. Whisper showed flat, minimal competitor activation throughout, suggesting it either does not represent lexical alternatives or suppresses them. We also examined a streaming variant of Whisper but observed no significant difference. From an engineering perspective, these results are not surprising: these models are trained to minimize Word Error Rate, an objective fundamentally different from simulating human online processing. CTC and encoder-decoder with attention are designed for transcription accuracy, not incremental competition, and thus encode fundamentally different recognition mechanisms. Rather than indicating failure, these results reveal that large-scale pretraining alone does not guarantee human-like incremental processing; architectural choices, learning objectives, and causal constraints shape processing strategies.

4. Limitations

Several limitations warrant consideration. First, our vocabulary is constrained to 1,533 words, whereas average adult lexicons likely contain 20,000 words or more, limiting ecological validity. Second, pretrained ASR models were trained on continuous speech, potentially biasing their evaluation on isolated words. This bias extends beyond task mismatch: Word frequency and neighborhood density—critical factors shaping lexical competition—are known to influence ASR inference, but remain uncontrolled in our current design [22]. Future work will address these factors using larger, carefully balanced lexicons and stimuli that better reflect naturalistic listening conditions.

5. Conclusions and future work

This work introduces a temporal benchmark to probe whether modern ASR models exhibit incremental word competition like humans. Using VWP metrics, we demonstrate that causality is a necessary architectural constraint for human-like dynamics: causal RNNs successfully replicate cohort suppression and rhyme activation, while most non-causal and foundation ASR models fail. These findings show that architectural choices, not scale alone, determine whether machines process speech incrementally in a human-compatible way. While large-scale models improve transcription accuracy, their failure to capture basic phonological competition warrants caution in interpreting them as models of human speech processing without careful behavioral validation. Future work will use these findings to design more biologically plausible architectures and validate them against continuous speech and neuroimaging data.

6. Acknowledgements

To preserve anonymity, acknowledgments are not included in this submission.

7. References

[1] P. D. Allopenna, J. S. Magnuson, and M. K. Tanenhaus, "Tracking the time course of spoken word recognition using eye movements: Evidence for continuous mapping models," *Journal of memory and language*, vol. 38, no. 4, pp. 419–439, 1998.

[2] W. D. Marslen-Wilson, "Functional parallelism in spoken word-recognition," *Cognition*, vol. 25, no. 1-2, pp. 71–102, 1987.

[3] J. L. McClelland and J. L. Elman, "The trace model of speech perception," *Cognitive psychology*, vol. 18, no. 1, pp. 1–86, 1986.

[4] J. S. Magnuson, H. You, S. Luthra, M. Li, H. Nam, M. Escabi, K. Brown, P. D. Allopenna, R. M. Theodore, N. Monto *et al.*, "Earshot: A minimal neural network model of incremental human speech recognition," *Cognitive science*, vol. 44, no. 4, p. e12823, 2020.

[5] J. Millet, C. Caucheteux, P. Orhan, Y. Boubenec, A. Gramfort, E. Dunbar, C. Pallier, and J.-R. King, "Toward a realistic model of speech processing in the brain with self-supervised learning," *Advances in Neural Information Processing Systems*, vol. 35, 2022.

[6] Y. Li, G. K. Anumanchipalli, A. Mohamed, P. Chen, L. H. Carney, J. Lu, J. Wu, and E. F. Chang, "Dissecting neural computations in the human auditory pathway using deep neural networks for speech," *Nature Neuroscience*, vol. 26, pp. 2213–2225, 2023.

[7] A. J. Anderson, C. Davis, and E. C. Lalor, "Deep-learning models reveal how context and listener attention shape the brain's response to speech," *PLOS Computational Biology*, 2024.

[8] O. Moussa and M. Toneva, "Brain-tuned speech models better reflect speech processing stages in the brain," in *Interspeech 2025*, 2025.

[9] M. Schrimpf, I. A. Blank, G. Tuckute, C. Kauf, E. A. Hosseini, N. Kanwisher, J. B. Tenenbaum, and E. Fedorenko, "The neural architecture of language: Integrative modeling converges on predictive processing," *Proceedings of the National Academy of Sciences*, vol. 118, no. 45, p. e2105646118, 2021.

[10] E. A. Hosseini, M. Schrimpf, Y. Zhang, S. Bowman, N. Zaslavsky, and E. Fedorenko, "Artificial neural network language models predict human brain responses to language even after a developmentally realistic amount of training," *Neurobiology of Language*, vol. 5, no. 1, pp. 43–63, 2024.

[11] M. Heilbron, K. Armeni, J.-M. Schoffelen, P. Hagoort, and F. P. de Lange, "A hierarchy of linguistic predictions during natural language comprehension," *Proceedings of the National Academy of Sciences*, vol. 119, no. 32, p. e2201968119, 2022.

[12] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural computation*, vol. 9, no. 8, pp. 1735–1780, 1997.

[13] Y. LeCun, Y. Bengio, and G. Hinton, "Deep learning," *nature*, vol. 521, no. 7553, pp. 436–444, 2015.

[14] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, "Attention is all you need," *Advances in neural information processing systems*, vol. 30, 2017.

[15] A. Graves, "Sequence transduction with recurrent neural networks," *arXiv preprint arXiv:1211.3711*, 2012.

[16] J. Kim, H. Lu, A. Tripathi, Q. Zhang, and H. Sak, "Reducing streaming asr model delay with self alignment," *Interspeech 2021*, pp. 3440–3444, 2021.

[17] J. Mahadeokar, Y. Shangguan, D. Le, G. Keren, H. Su, T. Le, C.-F. Yeh, C. Fuegen, and M. L. Seltzer, "Alignment restricted streaming recurrent neural network transducer," in *2021 IEEE Spoken Language Technology Workshop (SLT)*. IEEE, 2021, pp. 52–59.

[18] A. Graves, S. Fernández, F. Gomez, and J. Schmidhuber, "Connectionist temporal classification: labelling unsegmented sequence data with recurrent neural networks," in *Proceedings of the 23rd international conference on Machine learning*, 2006, pp. 369–376.

[19] A. Baevski, Y. Zhou, A. Mohamed, and M. Auli, "wav2vec 2.0: A framework for self-supervised learning of speech representations," *Advances in neural information processing systems*, vol. 33, pp. 12 449–12 460, 2020.

[20] W.-N. Hsu, B. Bolte, Y.-H. H. Tsai, K. Lakhotia, R. Salakhutdinov, and A. Mohamed, "Hubert: Self-supervised speech representation learning by masked prediction of hidden units," *IEEE/ACM transactions on audio, speech, and language processing*, vol. 29, pp. 3451–3460, 2021.

[21] J. S. Magnuson, "Trace-ing fixations in the visual world paradigm: Extending linking hypotheses and addressing individual differences by simulating trial-level behavior," *Brain Research*, vol. 1856, p. 149563, 2025.

[22] S. Goldwater, D. Jurafsky, and C. D. Manning, "Which words are hard to recognize? prosodic, lexical, and disfluency factors that increase speech recognition error rates," *Speech Communication*, vol. 52, no. 3, pp. 181–200, 2010.